

Theory of probability :-

Introduction:- If an experiment is repeated under essentially homogenous and similar conditions, we generally come across two type of situations.

1. The result or the outcome is unique or certain.
2. The result is not unique but may be one of the several possible outcome.

In tossing of a coin, one is not sure if head or tail will be obtained.

Similarly if a light tube has lasted for 't' hours, nothing can be said about its further life. In such cases we talk of chances or probability, which is taken to be a quantitative measure in certainty.

Trial and event :-

Consider an experiment which through repeated under essentially identical conditions doesn't give unique results but result may be in any one of the several possible outcomes. The experiment is known as a trial and the outcomes are known as events or cases.

For example:-

(i) Throwing of a dice is a trial and getting 1, 2, ... 6 is an event.

(ii) Tossing of a coin is a trial and getting head or tail is an event.

Exhaustive Events or cases:-

The total number of possible outcomes in any trial is known as exhaustive events or exhaustive cases

For example:- In tossing of a coin, there are two exhaustive cases, namely $\rightarrow$ head and tail.

~~In throwing of a coin, there are two exhaustive~~

In throwing of a dice there are six exhaustive cases namely 1, 2, 3, 4, 5, 6.

Favourable Events or cases:-

The number of cases favourable to an event in a trial is the number of outcomes which corresponds to the happening of the event.

For eg:-

(i) In drawing a card from a pack of 52 cards, the number of cases favourable to drawing of an ACE (A) is 4, for drawing a spade is 13, and a drawing a red card is 26.

(ii) In throwing of two dice, the number of cases favourable to getting the sum 5 is (1, 4), (4, 1), (2, 3), (3, 2) i.e. 4.

Mutually exclusive events or cases:-

Events are said to be mutually exclusive if the happening of any one of them removes the happening of all other possibilities. That means, no two or more of them can happen simultaneously in the same trial.

Independent events or cases:-

Several events are said to be independent if the happening of an event is not affected by the supplementary knowledge, concerning the occurrence of a no. of the remaining events. If we draw a card from a pack of 52 cards and replace it before drawing the second cards, the result of the second draw is independent of the first draw. But however if the first card drawn is not replaced then the second draw is dependent of the first draw.

Mathematical or classical definition of probabilities:-

If a trial results in 'n' no. of exhaustive mutually exclusive cases and 'm' no. of them are favourable to the happening of an event 'E' then the probability 'p' of happening of 'E' is given by

$$P = P(E) = \frac{\text{favourable no. of cases}}{\text{exhaustive no. of cases}} = \frac{m}{n}$$

Q) What is the chance that a leap year selected at random will contain 53 Sundays.

Soln:- In a leap year (which consists of 366 days) there are 52 complete weeks and two (2), extra days. The following are the possible combinations for this two extra days.

(Sunday and Monday), (Monday and Tuesday), (Tuesday and Wednesday), (Wednesday and Thursday), (Thursday and Friday), (Friday and Saturday), (Saturday and Sunday).

In other that a leap year selected at random should contain 53 sundays one of the two extra days must be sundays.

Since out of the above seven (7) possibilities only two (2) outcomes namely (SUN, MON), (SAT, SUN) satisfies the requirements.

Hence the required probabilities = $\frac{2}{7}$

Q A bag contains 3 red, 6 white and 7 blue balls. What is the probability that 2 balls drawn at random are white and blue?

Soln:-

Total number of balls = 16 (3+6+7)

Now, Out of 16 balls, 2 can be drawn in ${}^{16}C_2$ ways.

So, Exhaustive number of balls.

$${}^{16}C_2 = \frac{16!}{2! 14!} = \frac{16 \times 15 \times 14!}{2 \times 14!} \\ = 120$$

Out of 6 white ball can drawn is ${}^6C_1 = 6$ ways and similarly out of 7 blue balls, one ball can be drawn in 7C_1 ways.

Since each of the former cases can be associated with each of the latter cases, Therefore the total no. of

favourable cases:

$$= {}^6C_1 \times {}^7C_1$$

$$= \frac{6!}{5! \times 1} \times \frac{7!}{6! \times 1}$$

$$= 6 \times 7$$

$$= 42$$

Hence the required probability $= \frac{42}{120} = \frac{7}{20}$

Q Two cards are drawn at random from a pack of 52 cards show that the terms of drawing 2 ace is $\frac{1}{221}$.

Solⁿ:— From a pack of 52 cards two cards can be drawn in ${}^{52}C_2$ ways.

Hence the exhaustive no. of cards $= \frac{52!}{50! \cdot 2!}$

$$= \frac{52 \times 51 \times 50!}{50! \times 2}$$

=

Exhaustive no. of cards $= {}^{52}C_2$

In a pack, there are four aces and therefore two aces can be drawn in 4C_2 ways.

Hence req. probability $= \frac{{}^4C_2}{{}^{52}C_2} = \frac{1}{221}$

Q 4 cards are drawn from a pack of 52 cards. Find the probability that:-

- (i) All are diamond
- (ii) There is one card of suit.
- (iii) There are two spades and two hearts

Soln:-

Exhaustive no. of cases = ${}^{52}C_4$

$$= \frac{52 \times 51 \times 50 \times 49 \times 48!}{4! \times 48!}$$

$$= \frac{52 \times 51 \times 50 \times 49}{4 \times 3 \times 2 \times 1}$$

$$= 13 \times 17 \times 25 \times 49$$

(i) All are diamonds

No. of favourable cases = ${}^{13}C_4$

$$= 13 \times 11 \times 5$$

$$\text{Hence the req. probability} = \frac{13 \times 11 \times 5}{13 \times 17 \times 25 \times 49}$$

$$= \frac{11}{17 \times 5 \times 49}$$

(ii) There one one card of each suit

Each suit has 13 cards and one card from one suit is associated with the other, hence

$$\text{no. of favourable cases} = {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1$$

$$= 13 \times 13 \times 13 \times 13$$

$$\text{Req. probability} = \frac{(13)^4}{13 \times 17 \times 25 \times 49}$$

(iv) There are two spades - and two hearts.

Two spades can be selected in ${}^{13}C_2$ ways and two hearts can be ${}^{13}C_2$ ways. Since each of the way of selection of spade can be associated with each of the way of hearts, therefore the no. of favourable cases $= {}^{13}C_2 \times {}^{13}C_2$

$$\text{Req'd prob} = \frac{{}^{13}C_2 \times {}^{13}C_2}{13 \times 17 \times 49 \times 25}$$

8 If the letters of the word 'REGULATIONS' be arranged at random, what is the chance that there will be exactly four letters b/w R and E.

Soln:-

The word 'REGULATIONS' consist of 11 letters. 2 letters R and E can occupy ${}^{11}P_2 = \frac{11!}{9!} = 110$.

The no. of ways in which there will be exactly 4 letters b/w R and E, are listed below:-

- (i) R is in the 1st position & E is in the 6th postⁿ
- (ii) " " " " 2nd " " " " 7th "
- (iii) " " " " 3rd " " " " 8th "
- (iv) " " " " 4th " " " " 9th "
- (v) " " " " 5th " " " " 10th "
- (vi) " " " " 6th " " " " 11th "

Since R and E can interchange their positions, the req'd no. of favourable cases is 12.

$$\text{Hence the req'd probability} = \frac{12}{110}$$

5) A committee of 4 people, is to be appointed from 3 officers of the production department, four officers of the purchase department, two officers of the sales department and one chartered accountant. Find the probability of the forming the committee in the following manner.

i) There must be one from each category.

ii) It should have at least one from the purchase department.

iii) The C.A must be in the committee.

Soln.:-

There 10 officers in all and a committee of 4 people can be formed out of them in ${}^{10}C_4$ ways.

Hence,

$$\text{Exhaustive no. of cases} = {}^{10}C_4 = \frac{10!}{6!4!} = 210$$

(i) Favourable no. of cases for the committee to consist of four members one from each category is:

$${}^4C_1 \times {}^3C_1 \times {}^2C_1 \times {}^1C_1 = 4 \times 3 \times 2 \times 1 = 24$$

$$\text{Hence the reqd. probability is } \frac{24}{210} = \frac{12}{55}$$

(ii) $P(\text{committee has at least one purchase officer})$

$$= 1 - P(\text{committee has no. purchase officer})$$

In order that the committee has no purchase officer, All the four members are to be selected from amongst members of production sales and CA depts.
i.e. Out of the remaining 6 officers and 4 depts can

be done in 6C_4 ways.

$$= \frac{6!}{4! \times 2!}$$
$$= 15 \text{ ways.}$$

$$\therefore P(\text{Committee has no purchase officer}) = \frac{15}{210}$$
$$= \frac{1}{14}$$

$$\therefore P(\text{Committee has at least one purchase officer})$$
$$= 1 - \frac{15}{210} = 1 - \frac{1}{14} = \frac{13}{14}$$

iii) Since a chartered accountant can be selected out of 1 CA only in 1 way and remaining 3 members can be selected out of the remaining 9 officers in 9C_3 ways.

$$\text{Hence the reqd probability} = \frac{{}^9C_3}{210}$$

$$= \frac{9!}{3! \times 6!}$$
$$= \frac{84}{210}$$

$$= \frac{84}{210}$$

Probability function :-

$P(A)$ is the probability function, defined on the power set of sample space, if the following properties hold.

1. For each $A \in \mathcal{P}(S)$ $\} \# P: \mathcal{P}(S) \rightarrow (0,1)$
 $P(A)$ is defined, is real $\downarrow$
and $P(A) \geq 0$ (i.e. +ve) Sample space.

ii) For S , $P(S) = 1$, where S is the sample space

iii) If $\{A_n\}$ is any finite or infinite sequence of disjoint events in $\mathcal{P}(S)$, then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

☞ Show that $P(\emptyset) = 0$

Soln: $S \cup \emptyset = S$

$$P(S \cup \emptyset) = P(S)$$

$$P(S) + P(\emptyset) = P(S)$$

$$P(\emptyset) = 0$$

$\therefore$ Since S and $\emptyset$ are disjoint elements.

Theorem:-

Probability of complementary event $\bar{A}$ of A is given by

$$P(\bar{A}) = 1 - P(A)$$

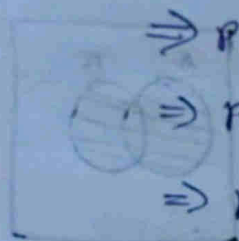
proof:-

A and $\bar{A}$ are disjoint event and $A \cup \bar{A} = S$

$$\Rightarrow P(A \cup \bar{A}) = P(S)$$

$$\Rightarrow P(A) + P(\bar{A}) = P(S) = 1$$

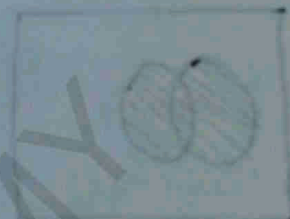
$$\Rightarrow P(\bar{A}) = 1 - P(A)$$



$A \cap \bar{A}$



A



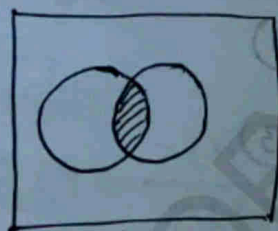
$A \cup B$

⑧ Thm:-

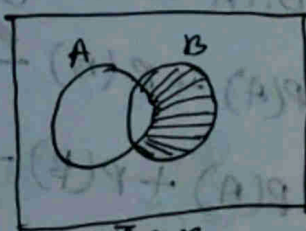
For any two events A and B

$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

proof:-



$A \cap B$



$\bar{A} \cap B$

From the above venn diagram we can see that the events $(A \cap B)$ and $(\bar{A} \cap B)$ are disjoint event and clearly

$$B = (A \cap B) \cup (\bar{A} \cap B)$$

$$P(B) = P((A \cap B) \cup (\bar{A} \cap B))$$

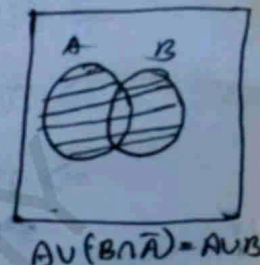
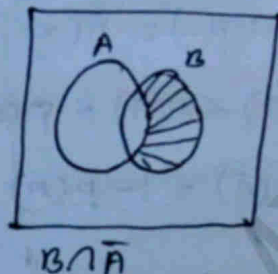
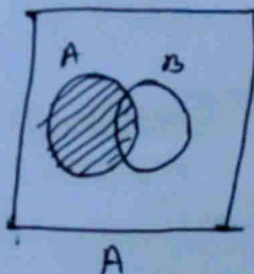
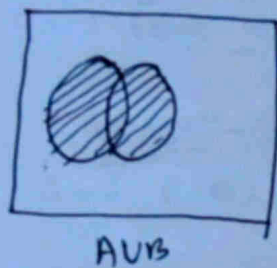
$$P(B) = P(A \cap B) + P(\bar{A} \cap B)$$

$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

Proof that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

proof:- $A \cup B$ can be written as the union of the two disjoint event A and $B \cap \bar{A}$



Hence,

$$A \cup B = A \cup (B \cap \bar{A})$$

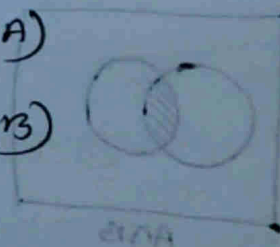
$$\Rightarrow P(A \cup B) = P(A \cup (B \cap \bar{A}))$$

$$\Rightarrow P(A \cup B) = P(A) + P(B \cap \bar{A})$$

$\therefore A$ and $B \cap \bar{A}$ are disjoint event

$$\Rightarrow P(A \cup B) = P(A) + P(B) - P(B \cap A)$$

$$\Rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Thm:- If $B \subset A$, then

$$(i) P(A \cap \bar{B}) = P(A) - P(B)$$

$$(ii) P(B) \leq P(A)$$

proof:- When $B \subset A$; B and $A \cap \bar{B}$ are disjoint event and their union is A

$$\therefore A = B \cup (A \cap \bar{B})$$

$$P(A) = P\{B \cup (A \cap \bar{B})\}$$

$$P(A) = P(B) + P(A \cap \bar{B})$$

$$\Rightarrow P(A) - P(B) = P(A \cap \bar{B})$$

(iv) We know that

$$P(A \cap B) \geq 0$$

$$\Rightarrow P(A) - P(B) \geq 0$$

$$\Rightarrow P(A) \geq P(B)$$

$$\Rightarrow P(B) \leq P(A)$$

Thm:- multiplication law of probability:-

For any two events A and B

$$P(A \cap B) = P(A) \cdot P(B/A) \text{ if } P(A) > 0$$

or

$$P(A \cap B) = P(B) \cdot P(A/B) \text{ if } P(B) > 0.$$

Where $P(B/A)$ represents the conditional probability of occurrence of which the event A has already happened and $P(A/B)$ is the conditional probability of happening of A, given that B has already happened.

proof:-

$$P(A) = \frac{n(A)}{n(S)}, \quad P(B) = \frac{n(B)}{n(S)}$$

$$\text{and } P(A \cap B) = \frac{n(A \cap B)}{n(S)} \quad \text{--- (i)}$$

For the conditional event A/B the favourable outcomes must be one of the sample points of B, i.e. for the event A/B the sample space B and out of the $n(B)$ sample point $n(A \cap B)$ pertain to the occurrence of the event A, Hence

$$P(A/B) = \frac{n(A \cap B)}{n(B)}$$

Rewriting eqn (i) we get

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{n(B)}{n(S)} \cdot \frac{n(A \cap B)}{n(B)} \\ = P(B) \cdot P(A/B)$$

Similarly we can prove that

$$P(A \cap B) = \frac{n(A)}{n(S)} \cdot \frac{n(A \cap B)}{n(A)} \\ = P(A) \cdot P(B|A)$$

This completes the proof of the equation.

Thm:- For any three events A, B and C

$$P(A \cup B | C) = P(A|C) + P(B|C) - P(A \cap B | C)$$

proof:- we know that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\therefore P\{(A \cap C) \cup (B \cap C)\} = P(A \cap C) + P(B \cap C) - P\{(A \cap C) \cap (B \cap C)\} \\ = P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$$

Dividing b/s by $P(C)$; we get,

$$\frac{P\{(A \cap C) \cup (B \cap C)\}}{P(C)} = \frac{P(A \cap C)}{P(C)} + \frac{P(B \cap C)}{P(C)} - \frac{P(A \cap B \cap C)}{P(C)}$$

$$\Rightarrow \frac{P\{(A \cup B) \cap C\}}{P(C)} = \frac{P(A \cap C)}{P(C)} + \frac{P(B \cap C)}{P(C)} - \frac{P(A \cap B \cap C)}{P(C)}$$

$$\Rightarrow \frac{P\{(A \cup B) \cap C\}}{P(C)} = P(A|C) + P(B|C) - P(A \cap B | C)$$

Thm:- For any three event A, B and C

$$P(A \cap \bar{B} | C) + P(A \cap B | C) = P(A | C)$$

Proof:-

$$P(A \cap \bar{B} | C) + P(A \cap B | C) \\ = \frac{P(A \cap \bar{B} \cap C)}{P(C)} + \frac{P(A \cap B \cap C)}{P(C)}$$

$$= \frac{P(A \cap \bar{B} \cap C) + P(A \cap B \cap C)}{P(C)}$$

Now since $A \cap \bar{B} \cap C$ and $A \cap B \cap C$ are disjoint events and it can be easily prove that $A \cap C = (A \cap \bar{B} \cap C) \cup (A \cap B \cap C)$

$$\Rightarrow P(A \cap C) = P\{(A \cap \bar{B} \cap C) \cup (A \cap B \cap C)\}$$

$$\Rightarrow P(A \cap C) = P(A \cap \bar{B} \cap C) + P(A \cap B \cap C) \quad \text{--- (**)}$$

Now by using eqn (*) and (**) we get

$$P(A \cap \bar{B} / C) + P(A \cap B / C) = \frac{P(A \cap C)}{P(C)}$$

$$\Rightarrow P(A \cap \bar{B} / C) + P(A \cap B / C) = P(A / C)$$

Independent event :-

An event B is said to be independent to another event A if

$$P(A \cap B) = P(A) \cdot P(B)$$

Thm:- If A and B are independent event then $\bar{A}$ and $\bar{B}$ are also independent event.

Since A and B are independent event, $\therefore$

$$P(A \cap B) = P(A) \cdot P(B)$$

$$\text{Now } P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) \quad \therefore \bar{A} \cap \bar{B} = \overline{A \cup B}$$

$$\Rightarrow P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B)$$

$$= 1 - \{P(A) + P(B) - P(A \cap B)\}$$

$$= 1 - \{P(A) + P(B) - P(A) \cdot P(B)\}$$

$$= \{1 - P(B)\} - P(A) \{1 - P(B)\}$$

$$\Rightarrow (1 - P(B))(1 - P(A)) = P(\bar{B}) \cdot P(\bar{A})$$

$$= P(\bar{A}) \cdot P(\bar{B})$$

$$\Rightarrow P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B}) = P(\bar{A} \cap \bar{B})$$

Hence $\bar{A}$ and $\bar{B}$ are independent events.

Q. If two dice are thrown what is the probability that the sum is

- (i) Greater than 8 (ii) Neither 7 nor 11

Ans: (i) If S denotes the sum of the two dice then we want $P(S > 8)$

The required event can happen in the following mutually exclusive ways

$$S=9, S=10, S=11, S=12$$

$$\therefore P(S > 8) = P(S=9 \cup S=10 \cup S=11 \cup S=12)$$

$$= P(S=9) + P(S=10) + P(S=11) + P(S=12)$$

In throw a two dice in a sample space contains 36 ways

$$S=9 : (3,6), (6,3), (4,5), (5,4)$$

$$\therefore P(S=9) = \frac{4}{36} = \frac{1}{9}$$

$$P(S=10) = \frac{3}{36} = \frac{1}{12}, \quad P(S=11) = \frac{2}{36} = \frac{1}{18}$$

$$P(S=12) = \frac{1}{36}$$

$$\therefore \text{Therefore } P(S > 8) = \frac{1}{9} + \frac{1}{12} + \frac{1}{18} + \frac{1}{36} = \frac{10}{36} = \frac{5}{18}$$

(ii) Let A denote the event of getting the sum of 7 and B denote the event of getting the sum of 11 with a pair of dice.

$$S: (1,6), (6,1), (2,5), (5,2), (3,4), (4,3)$$

$$\therefore P(S=7) = \frac{6}{36} = \frac{1}{6}$$

$$P(S=11) = \frac{2}{36} = \frac{1}{18}$$

$$\text{Required probability} = P(\bar{A} \cap \bar{B})$$

$$= P(\overline{A \cup B})$$

$$= 1 - P(A \cup B)$$

$$= 1 - P(A \cup B)$$

$$= 1 - \{P(A) + P(B)\}$$

$$= 1 - \left(\frac{1}{6} + \frac{1}{18}\right)$$

8 A problem in statistics is given to the three students A, B and C whose chances of solving it are $\frac{1}{2}$, $\frac{3}{4}$ and $\frac{1}{4}$ respectively. What is the probability that the problem will be solved if all of them try independently.

Ans: Let A, B and C denote the event that the problem is solved by the student A, B, C respectively

$$\text{then, } P(A) = \frac{1}{2}, P(B) = \frac{3}{4} \text{ and } P(C) = \frac{1}{4}$$

When the problem will be solved if at least one of them solves the problem. Then we have to calculate the probability of occurrence of at least one of the three events A, B and C.

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

$$= P(A) + P(B) + P(C) - P(A) \cdot P(B) - P(B) \cdot P(C) - P(C) \cdot P(A) +$$

$$P(A) \cdot P(B) \cdot P(C)$$

$$= \frac{1}{2} + \frac{3}{4} + \frac{1}{4} - \left(\frac{1}{2} \cdot \frac{3}{4}\right) - \left(\frac{3}{4} \cdot \frac{1}{4}\right) - \left(\frac{1}{4} \cdot \frac{1}{2}\right) - \left(\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4}\right)$$

$$= \frac{1}{2} + \frac{3}{4} + \frac{1}{4} - \frac{8}{32} - \frac{3}{16} - \frac{1}{8} - \frac{3}{32}$$

$$=$$

⑧ The odds against manager X, solving the ways dispute with the workers are 8:6 and odds in favour of manager Y solving the same dispute are 14:16

(i) What is the chance that neither solves the dispute if they both try independently each other.

(ii) What is the probability that the dispute will be solved.

Ans:-

Let A be the event that the manager X, will solve the dispute and B be the event the manager Y will solve the dispute.

then clearly,

$$P(\bar{A}) = \frac{8}{14}$$

$$P(A) = \frac{6}{14}$$

$$\text{and } P(B) = \frac{14}{30}, P(\bar{B}) = \frac{16}{30}$$

The reqd probability that neither settles the dispute is given by

$$P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B})$$

$$= \frac{8}{14} \cdot \frac{16}{30}$$

$$= \frac{32}{105}$$

(ii) The dispute will be settled if at least one of the managers X and Y settles the dispute.

Hence, reqd probability given by

$$P(A \cup B) = 1 - P(\bar{A} \cap \bar{B})$$

$$= 1 - P(\bar{A} \cap \bar{B})$$

$$= 1 - \frac{32}{105} = \frac{73}{105}$$

⑧ The odds that person X speaks the truth are 3:2 and the odds that person Y speaks the truth are 5:3. In what percentage of cases are they likely to contradict each other on an identical point.

Ans: — let us define the events

A: X speaks the truth

B: Y speaks the truth.

then $\bar{A}$ and $\bar{B}$ represents the complementary events that X and Y tell a lie respectively.

$$P(A) = \frac{3}{3+2} = \frac{3}{5}, \quad P(\bar{A}) = 1 - \frac{3}{5} = \frac{2}{5}$$

$$P(B) = \frac{5}{5+3} = \frac{5}{8}, \quad P(\bar{B}) = \frac{3}{8}$$

The event E that X and Y contradict each other on a identical point can happen in the following exclusive ways.

(i) X speaks the truth and Y tells a lie;

i.e. the event $A \cap \bar{B}$ happens

(ii) X tells a lie and Y speaks the truth

i.e. the event $\bar{A} \cap B$ happens

Hence by addition theorem of probability the required probability is given by:-

$$P(E) = P(A \cap \bar{B}) + P(\bar{A} \cap B) \quad \text{--- (*)}$$

Now Since A and B are independent events so A and $\bar{B}$ will be independence events and similarly $\bar{A}$ and B will also independent event.

Therefore,

$$P(A \cap \bar{B}) = P(A) \cdot P(\bar{B})$$

$$\& P(\bar{A} \cap B) = P(\bar{A}) \cdot P(B)$$

Therefore from equation no. (4) we get

$$P(E) = P(A) \cdot P(\bar{B}) + P(\bar{A}) \cdot P(B)$$

$$= \frac{3}{5} \times \frac{3}{8} + \frac{2}{5} \times \frac{5}{8}$$

$$= \frac{19}{40}$$

$$= 0.475$$

Hence A and B are likely to contradict to each other on an identical point in 47.5% of the cases.

Unit - II

Random variable:-

In general by a random variable we mean a function x connected with the outcome of a random experiment E .

for example:-

If E consist of 2 tosses of a coin we may contain the random variable which is the numbers of heads.

Outcome:	HH	HT	TH	TT
Value of x :	2	1	1	0

Thus to each outcome ω , there corresponds a real number $x(\omega)$. Since the points of the sample space S corresponds to the outcomes of an experiment that means a real number, which is denoted by

$x(\omega)$, is related with every $\omega \in S$.

Let S be the sample space associated with a given random experiment. Then a real valued function defined on S and taking values in $\mathbb{R} (-\infty, \infty)$ is called a one dimensional random variable.

Note:- We shall need to make probability statements about a random variable x , such as $(x \leq a)$. For the example given above we should write $\{P(x \leq 1) = P\{TT, TH, HT\} = \frac{3}{4}$. That is $P(x \leq a)$ is simply the probability of the set of outcomes ω for which $x(\omega) \leq a$ or $P(x \leq a) = P\{\omega \in S: x(\omega) \leq a\}$

Distribution function:- Let x be a random variable. Then the function,

$$F_x(x) = P(x \leq x) \\ = P\{\omega \in S: x(\omega) \leq x\}$$

is called the distribution function of x . In general we write $F(x)$ instead of $F_x(x)$.

Properties of distribution function:-

(i) If F is the distribution function of the random variable x and if $a < b$ then $P(a < x \leq b) = F(b) - F(a)$

Proof:-

The event ' $a < x \leq b$ ' and $x \leq a$ are disjoint and their union is the event ' $x \leq b$ '. Hence by addition theorem of probability.

$$P(a < x \leq b) + P(x \leq a) = P(x \leq b)$$

$$\Rightarrow P(a < x \leq b) = P(x \leq b) - P(x \leq a)$$

$$P(a < x \leq b) = F(b) - F(a)$$

$$\left[\begin{aligned} P(x \leq b) \\ &= P\{x \leq a\} \cup (a < x \leq b) \\ &= P(x \leq a) + P(a < x \leq b) \end{aligned} \right]$$

(ii) If F is the distribution function of one dimensional random variable x , then,

$$0 \leq F(x) \leq 1 \text{ and } F(x) < F(y) \text{ if } x < y$$

$$F(x) = P(x \leq x)$$

$$F(y) = P(x \leq y)$$

$$F(y) = P\{ \dots \cup \dots$$

$$= P(x \leq x) + P(x < x \leq y)$$

$$= F(x) + (\text{Some true } \in)$$

Discrete random variable:-

If a random variable a most countable number of values, it is called a discrete random variable. In other words, a real value function defined on a discrete sample space called a discrete random variable.

Probability mass function:- Suppose x is a one dimensional discrete random variable taking of most a countable number of values $x_1, x_2, x_3 \dots$ with each possible outcome x_i , we associate a number $p_i = P(x = x_i) = P(x_i)$ is called the probability of x_i . The numbers $P(x_i)$, $i = 0, 1, 2, 3, \dots$ must satisfy the following condition.

$$(i) P(x_i) \geq 0, \forall i \quad (ii) \sum_{i=1}^{\infty} P(x_i) = 1$$

This function P is called the probability mass function of the random variable x , and the set $\{x_i, P(x_i)\}$ is called the probability distribution of random variable x .

§ A random variable x on the following probability distribution:-

x :	0	1	2	3	4	5	6	7
$P(x=x_i)$:	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$9k^2 + k$

(i) Find k

(ii) Evaluate $P(x < 6)$, $P(x \geq 6)$ and $P(0 < x < 5)$

(iii) If $P(x \leq c) > \frac{1}{2}$, find the minimum value of c

and (iv) Determine the distribution function of x

Ans:-

(i) Since P is a probability mass function and hence $\sum_{x=0}^7 P(x=x_i) = 1$

$$\Rightarrow 0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 9k^2 + k = 1$$

$$\Rightarrow 10k^2 + 9k - 1 = 0$$

$$\Rightarrow (10k-1)(k+1) = 0$$

$$\Rightarrow k = \frac{1}{10} \text{ or } k = -1$$

Now since probability can not be negative, $k = -1$ is rejected and hence $k = \frac{1}{10}$

$$(ii) P(x < 6) = P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4) + P(x=5)$$

$$= 0 + k + 2k + 2k + 3k + k^2$$

$$= 8k + k^2$$

$$= \frac{8}{10} + \frac{1}{100} = \frac{81}{100} \quad \left[\because k = \frac{1}{10} \right]$$

$$P(X \geq 6) = 1 - P(X \leq 5) = 1 - \frac{81}{100} = \frac{19}{100}$$

$$P(0 \leq X \leq 5) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= k + 2k + 2k + 3k + 3k + 3k$$

$$= 8k$$

$$= \frac{8}{10}$$

(iii) If $c=0$ then $P(X \leq 0) = 0$, does not satisfy the given condition.

if $c=1$; then $P(X \leq 1)$

$$= P(X=0) + P(X=1)$$

$$= 0 + k = \frac{1}{10} \neq \frac{1}{2}$$

if $c=2$, then $P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$$= 0 + k + 2k = 3k = \frac{3}{10} \neq \frac{1}{2}$$

if $c=3$, then $P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$

$$= 0 + k + 2k + 2k = 5k = \frac{5}{10} = \frac{1}{2} \neq \frac{1}{2}$$

if $c=4$, then, $P(X \leq 4) = \{P(X=0) + P(X=1) + P(X=2) + P(X=3)\} + P(X=4)$

$$= \frac{1}{2} + 3k = \frac{1}{2} + \frac{3}{10} = \frac{8}{10} > \frac{1}{2}$$

So the minimum value of c for which

$$P(X \leq c) \geq \frac{1}{2} \text{ is } c=4$$

Hence $c=4$

(iv)

$$X \mid F_X(x) = P(X \leq x)$$

0

$$P(X \leq 0) = 0$$

1

$$P(X \leq 1) = k = \frac{1}{10}$$

2

$$P(X \leq 2) = \frac{3}{10}$$

3

$$P(X \leq 3) = \frac{1}{2}$$

4

$$P(X \leq 4) = \frac{8}{10}$$

5

$$P(X \leq 5) = \frac{8}{10} + \frac{1}{10} = \frac{9}{10}$$

6

$$P(X \leq 6) = \frac{9}{10} + \frac{2}{10} = \frac{11}{10}$$

7

$$P(X \leq 7) = \frac{9}{10} + \frac{7}{10} + \frac{1}{10} = 1$$

5) Let x be a random variable such that

$$P(x=2) = P(x=-1),$$

$$P(x=2) = P(x=1)$$

$$\text{and, } P(x>0) = P(x\geq 0) = P(x=0)$$

obtain the probability mass function of x and its distribution function.

Ans:-

$$P(x>0) = P(x=1) + P(x=2)$$

$$P(x<0) = P(x=-1) + P(x=-2)$$

Let, $P(x=-2) = t$, Hence as per question

$$P(x=-1) = t$$

$$\therefore P(x<0) = P(x=-1) + P(x=-2)$$

$$= t + t$$

$$= 2t$$

$$\text{Again, } P(x>0) = P(x\geq 0) = 2t$$

$$\text{and } P(x=0) = P(x<0) = 2t$$

Since, p is a probability function, hence

$$P(x=0) + P(x<0) + P(x>0) = 1$$

$$\Rightarrow 2t + 2t + 2t = 1$$

$$\Rightarrow t = \frac{1}{6}$$

$$\therefore P(x=-2) = P(x=-1) = \frac{1}{6}$$

$$P(x=0) = 2t = \frac{2}{6} = \frac{1}{3}$$

$$P(x>0) = 2t$$

$$\Rightarrow P(x=1) + P(x=2) = 2t$$

$$\Rightarrow P(x=1) = t = \frac{1}{6}$$

($\because P(x=1) = P(x=2)$ as per question)

$$\Rightarrow P(X=2) = t = \frac{1}{6}$$

Thus the distribution function is as given below:-

X	$P(X=x)$	$F(x) = P(X \leq x)$
-2	$\frac{1}{6}$	$\frac{1}{6}$
-1	$\frac{1}{6}$	$\frac{2}{6}$
0	$\frac{2}{6}$	$\frac{4}{6}$
1	$\frac{1}{6}$	$\frac{5}{6}$
2	$\frac{1}{6}$	$\frac{6}{6} = 1$

8 A random variable X assumes the values $-3, -2, -1, 0, 1, 2, 3$ such that

$$P(X=-3) = P(X=-2) = P(X=-1)$$

$$P(X=1) = P(X=2) = P(X=3)$$

$$\text{and } P(X=0) = P(X < 0) = P(X > 0)$$

Obtain the probability mass function of X and distribution function.

Ans:-

$$\text{Let } P(X=-3) = t, \text{ A/g}$$

$$\text{A/g } P(X=-2) = P(X=-1)$$

$$P(X < 0) = P(X=-3) + P(X=-2) + P(X=-1) = 3t$$

$$\therefore P(X > 0) = P(X < 0) \quad \text{A/g}$$

$$= 3t$$

Similarly A/g

$$P(X=0) = P(X < 0) = 3t$$

Since, P is a probability mass function.

$$P(X < 0) + P(X = 0) + P(X > 0) = 1$$

$$\Rightarrow 3t + 3t + 3t = 1$$

$$\Rightarrow t = \frac{1}{9}$$

$$\therefore P(X = 0) = 3t = \frac{3}{9} = \frac{1}{3}$$

$$\therefore P(X = -3) = P(X = -2) = P(X = -1) = t = \frac{1}{9}$$

Now,

$$P(X > 0) = P(X = 1) = 3t = \frac{3}{9} = \frac{1}{3} \quad \text{A(9)}$$

$$\Rightarrow P(X = 1) + P(X = 2) + P(X = 3) = \frac{1}{3}$$

$$\Rightarrow 3P(X = 1) = \frac{1}{3} \quad \left(\text{A(9)} \quad P(X = 1) = P(X = 2) = P(X = 3) \right)$$

$$\Rightarrow P(X = 1) = \frac{1}{9}$$

$$\Rightarrow P(X = 2) = P(X = 3) = \frac{1}{9}$$

The distribution function is

X	$P(X = x)$	$F(x) = P(X \leq x)$
-3	$\frac{1}{9}$	$\frac{1}{9}$
-2	$\frac{1}{9}$	$\frac{2}{9}$
-1	$\frac{1}{9}$	$\frac{3}{9}$
0	$\frac{1}{3}$	$\frac{4}{9}$
1	$\frac{1}{9}$	$\frac{5}{9}$
2	$\frac{1}{9}$	$\frac{6}{9}$
3	$\frac{1}{9}$	$\frac{7}{9} = 1$

22 Aug

Q A random variable has the following probability distribution

$X:$	0	1	2	3	4	5	6	7	8
$P(X=x):$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

(i) Determine the value of a

(ii) Find $P(X < 3)$, $P(X \geq 3)$, $P(0 < X < 5)$

(iii) What is the smallest value of x for which $P(X \leq x) > 0.5$

(iv) Find out the distribution function of X .

Ans:— Since p is a probability mass function, hence

$$\sum_{x=0}^8 P(X=x) = 1$$

$$\Rightarrow a + 3a + 5a + 7a + 9a + 11a + 13a + 15a + 17a = 1$$

$$\Rightarrow 81a = 1$$

$$\Rightarrow a = \frac{1}{81}$$

$$(ii) P(X < 3) = P(X=0) + P(X=1) + P(X=2)$$

$$= a + 3a + 5a = 9a = \frac{9}{81}$$

$$P(X \geq 3) = 1 - P(X < 3)$$

$$= 1 - \frac{9}{81}$$

$$= \frac{72}{81}$$

$$P(0 < X < 5) = P(X=1) + P(X=2) + P(X=3) + P(X=4)$$

$$= 3a + 5a + 7a + 9a$$

$$= 24a$$

$$= \frac{24}{81}$$

$$(iv) P(X \leq 0) = P(X=0) = \frac{1}{81} < 0.5$$

$$P(X \leq 1) = P(X=0) + P(X=1) \\ = \frac{4}{81} < 0.5$$

$$P(X \leq 2) =$$

$$P(X \leq 3) =$$

$$P(X \leq 4) =$$

$$P(X \leq 5) = P(X=0) + \dots + P(X=5) \\ = \frac{36}{81} < 0.5$$

$$P(X \leq 6) = P(X=0) + \dots + P(X=6) \\ = \frac{49}{81} > 0.5$$

Hence the smallest value of x for which

$$P(X \leq x) > 0.5 \text{ is } 6$$

iv Distribution function of X

X	$F_X(x) = P(X \leq x)$
0	$\rightarrow \frac{1}{81}$
1	$\rightarrow \frac{4}{81}$
2	$\rightarrow \frac{1}{9}$
3	$\rightarrow \frac{14}{81}$
4	$\rightarrow \frac{25}{81}$
5	$\rightarrow \frac{36}{81}$
6	$\rightarrow \frac{49}{81}$
7	$\rightarrow \frac{64}{81}$
8	$\rightarrow \frac{81}{81} = 1$

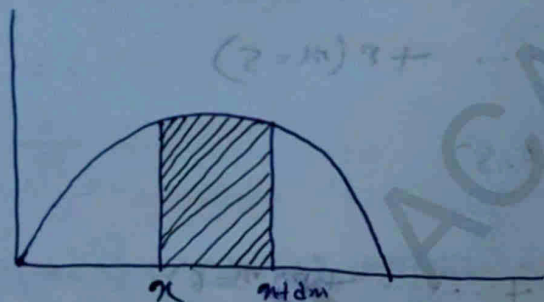
Continuous random variable

A random variable x is said to be continuous if it can take all possible values between certain limits.

Probability density function:-

Consider the small interval $(x, x+dx)$ of length dx round the point x . Let $f(x)$ be any continuous function of x so that $f(x)dx$ represents the probability that x falls in the interval $(x, x+dx)$: Symbolically

$$P(x \leq X \leq x+dx) = f(x) dx$$



In the above figure represent the area bounded by the curve $y = f(x)$, x -axis and the ordinates at the point x and $x+dx$. The function $f(x)$; so defined is known as probability density f^x or probability density function.

Defⁿ:- The probability density function $f_x(x)$ of random variable x is defined as $f_x(x) = \lim_{\delta x \rightarrow 0} \frac{P(x \leq X \leq x + \delta x)}{\delta x}$

The probability for a variable to lie in the interval dx is $f(x).dx$ and hence the probability for a variable x falls in the interval $[a, b]$ is P

$$P(a \leq x \leq b) = \int_a^b f(x) dx$$

Note:- The probability density function of a random variable X , usually denoted by $f_X(x)$ or simply by $f(x)$ has the following obvious properties.

(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

Q The diameter of an electric cable say x is assume to be a continuous random variable with probability density function.

$$f(x) = 6x(1-x), 0 \leq x \leq 1$$

(i) Check that above is probability density function.

(ii) Determine a member 'b' such that

$$P(x < b) = P(x > b)$$

Ans:-

Hence,

$$f(x) \geq 0, (\because 0 \leq x \leq 1)$$

$$\text{Now, } \int_{-\infty}^{\infty} f(x) dx = \int_0^1 f(x) dx$$

$$= \int_0^1 6x(1-x) dx$$

$$= 6 \int_0^1 (x - x^2) dx$$

$$= 6 \left\{ \frac{x^2}{2} - \frac{x^3}{3} \right\}$$

$$= 6 \left\{ \frac{1}{2} - \frac{1}{3} \right\}$$

$$= 6 \times \frac{1}{6}$$

$$= 1$$

$$(1) P(x < b) = P(x > b)$$

$$\Rightarrow \int_0^b f(m) dm = \int_b^1 f(m) dm$$

$$\Rightarrow 6 \int_0^b x(1-x) dx = 6 \int_b^1 x(1-x) dx$$

$$\Rightarrow \left(\frac{x^2}{2} - \frac{x^3}{3} \right)_0^b = \left(\frac{x^2}{2} - \frac{x^3}{3} \right)_b^1$$

$$\Rightarrow \frac{b^2}{2} - \frac{b^3}{3} = \left(\frac{1}{2} - \frac{1}{3} \right) - \left(\frac{b^2}{2} - \frac{b^3}{3} \right)$$

$$\Rightarrow 4b^3 - (b^2 + 1) = 0$$

$$\Rightarrow (2b-1)(2b^2-2b-1) = 0$$

$$\Rightarrow b = \frac{1}{2} \text{ or } 2b^2 - 2b - 1 = 0$$

doesn't have any real solⁿ

$$\text{Hence } b = \frac{1}{2}$$

⊗ Let x be a continuous random variable with p.d.f $f(m)$ given by

$$f(m) = \begin{cases} km, & 0 \leq x < 1 \\ k, & 1 \leq x < 2 \\ -km + 3k, & 2 \leq x < 3 \\ 0, & x \geq 3 \end{cases}$$

(i) Determine k

(ii) Determine $f(m)$

Ans:— Since $f(m)$ is the p.d.f

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^0 f(m) dm + \int_0^1 f(m) dm + \int_1^2 f(m) dm + \int_2^3 f(m) dm + \int_3^{\infty} f(m) dm = 1$$

$$\Rightarrow 0 + \int_0^1 k n \, dn + \int_1^2 k \, dn + \int_2^3 (-k n + 3k) \, dn + 0 = 1$$

$$\Rightarrow 0 + \frac{k n^2}{2} \Big|_0^1 + k n \Big|_1^2 + \left\{ -\frac{k n^2}{2} + 3k n \right\}_2^3 = 1$$

$$\Rightarrow \frac{k}{2} \left(\frac{1}{2} - 0 \right) + k(2-1) + k \left(\frac{9}{2} - \frac{4}{2} \right) + 3k(3-2) = 1$$

$$\Rightarrow \frac{k}{2} + k - \frac{9k}{2} + 2k + 3k = 1$$

$$\Rightarrow 6k + \frac{k}{2} - \frac{9k}{2} = 1$$

$$\Rightarrow \frac{12k + k - 9k}{2} = 1$$

$$\Rightarrow \frac{4k}{2} = 1$$

$$\Rightarrow 2k = 1$$

$$\Rightarrow k = \frac{1}{2}$$

$$(ii) \text{ If } -\alpha < n < 0$$

$$\text{then } f(n) = 0$$

Now if $0 \leq n < 1$ then

$$F(n) = \int_0^1 f(n) \, dn = \int_0^1 k n \, dn = k \frac{n^2}{2} \Big|_0^1$$

$$= \frac{k}{2} = \frac{1}{4}$$

Now if $1 \leq n < 2$ then

$$f(n) = \int_0^2 f(n) \, dn = \int_0^1 f(n) \, dn + \int_1^2 f(n) \, dn$$

=

Binomial distribution :-

Binomial distribution was discovered by James Bernoulli in the year 1700. Let random experiment be performed repeatedly, each repetition being trial and let the occurrence of an event in a trial be called as success and its non occurrence is failure.

Consider a set of n independent trials in which the probability ' p ' of success in any trial is constant and then the probability of failure ($q = 1 - p$)

The probability of x success and consequently $(n-x)$ failures in ' n ' independent trials, in a special order.

SSFSFFFS ... FSF

where S represents a success and F represents the failure.

Now,

$$\begin{aligned} & P(SSFSFFFS \dots FSF) \\ &= P(S) \cdot P(S) \cdot P(F) \cdot P(S) \cdot P(F) \cdot P(F) \cdot P(F) \cdot P(S) \dots P(F) \cdot P(S) \cdot P(F) \\ &= p \cdot p \cdot q \cdot p \cdot q \cdot q \cdot q \dots q \cdot p \cdot q \end{aligned}$$

Let the no. of success $= x$

no. of failure $= n - x$

Hence, we get from the above

$$= p^x \cdot q^{n-x}$$

But x no. of success out of n no. of trials can occur in nC_x ways and the probabilities for each of these ways is saying that means $p^x \cdot q^{n-x}$

Hence the probability of x no. of success in n no. of trials in any order is given by the probability expression ${}^n C_x \cdot p^n \cdot q^{n-x}$

Defⁿ:- A random variable x is said to follow Binomial Distribution if it assumes only non-negative values and its probability mass function is given by

$$P(X=x) = \begin{cases} {}^n C_x \cdot p^n \cdot q^{n-x}; & x=0, 1, 2, \dots, n \\ 0, & \text{otherwise.} \end{cases}$$

Q. 10 coins are thrown simultaneously. Find the probability of getting atleast 7 head.

Ans:-

Let P = Probability of getting a head = $\frac{1}{2}$

q = Probability of getting a tail = $\frac{1}{2}$

The probability of obtaining x no. of head in a random 10 coins

$$P(x) = {}^{10} C_x \left(\frac{1}{2}\right)^x \cdot \left(\frac{1}{2}\right)^{10-x} \quad \text{--- (i)}$$

Probability of getting atleast 7 head in a random throw of 10 coin given by

$$P(X \geq 7)$$

$$= P(X=7) + P(X=8) + P(X=9) + P(X=10)$$

$$= {}^{10} C_7 \left(\frac{1}{2}\right)^7 \cdot \left(\frac{1}{2}\right)^3 + {}^{10} C_8 \left(\frac{1}{2}\right)^8 \cdot \left(\frac{1}{2}\right)^2 + {}^{10} C_9 \left(\frac{1}{2}\right)^9 \cdot \left(\frac{1}{2}\right) + {}^{10} C_{10} \left(\frac{1}{2}\right)^{10}$$

$$= \frac{1}{2^{10}} \{ {}^{10} C_7 + {}^{10} C_8 + {}^{10} C_9 + {}^{10} C_{10} \}$$

$$= \frac{1}{1024} \left\{ \frac{10!}{7! \times 3} + \frac{10!}{8! \times 2} + \frac{10!}{9! \times 1} + 1 \right\}$$

$$= \frac{1}{1024} \left\{ \frac{10 \times 9 \times 8}{6} + \frac{10 \times 9}{2} + 10 + 1 \right\}$$

$$= \frac{1}{1024} \{ 120 + 45 + 11 \}$$

$$= \frac{176}{1024}$$

12 Sept 2025

Q A and B play a game in which their chances of winning are in the ratio 3:2. Find A's chance of winning at least three games out of five games played.

Ans:- Let p be the probability that 'A' wins the game,

Then, $n=5, p=\frac{3}{5}, q=\frac{2}{5}$

Hence by binomial probability law, the probability that out of 5 games played, A wins ' x ' no. of games by:

$$P(X=x) = {}^5C_x \left(\frac{3}{5}\right)^x \left(\frac{2}{5}\right)^{5-x}, \quad x=0,1,2,3,4,5.$$

The req. probability that out of five games A wins at least three games is given by:-

$$P(X \geq 3) = P(X=3) + P(X=4) + P(X=5)$$

$$= {}^5C_3 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right)^2 + {}^5C_4 \left(\frac{3}{5}\right)^4 \left(\frac{2}{5}\right) + {}^5C_5 \left(\frac{3}{5}\right)^5 \left(\frac{2}{5}\right)^0$$

$$= \frac{3^3}{5^3} \left\{ \frac{5!}{3! \times 2!} \cdot \frac{4}{25} + \frac{5!}{4!} \cdot \frac{3 \times 2}{5 \times 5} + \frac{0}{25} \right\}$$

$$= \frac{27}{125} \left\{ \frac{10 \times 4}{25} + \frac{5 \times 3 \times 2}{5 \times 5} + \frac{0}{25} \right\}$$

$$= \frac{27}{125} \left\{ \frac{40 + 30 + 0}{25} \right\} = \frac{27 \times 70}{125 \times 25}$$
$$= 0.68$$

Q) A multiple choice test consist of 8 question with each 3 answers to which question?

The student answer each question by throwing a dice and choose the first option if he gets 1 or 2, The second option if he gets 3 or 4 and the last option if he get 5 or 6. To get a distinction, student must secure atleast 75% correct answer. If there is no negative marking, what is the prob. that the student secures the distinction?

Ans

Since there are 3 answers to each question out of which only one is correct, the prob. of getting an answer to a question correctly is

$$p = \frac{1}{3}, q = \frac{2}{3}$$

By binomial distribution the prob. of getting r no. of correct answers in a '8' question test is given by

$$P(X=r) = {}^8C_r \left(\frac{1}{3}\right)^r \cdot \left(\frac{2}{3}\right)^{8-r} \quad \text{where, } r=0, 1, 2, 3, \dots$$

We get a distinction the student must get atleast 6 question answer correctly atleast 6 question

$$P(n \geq 6) = P(n=6) + P(n=7) + P(n=8)$$

$$= {}^8C_6 \left(\frac{1}{3}\right)^6 \cdot \left(\frac{2}{3}\right)^2 + {}^8C_7 \left(\frac{1}{3}\right)^7 \left(\frac{2}{3}\right) + {}^8C_8 \left(\frac{1}{3}\right)^8 \left(\frac{2}{3}\right)^0$$

$$= \frac{128}{729 \times 9}$$

Q. A Department in a work has 10 machines which may need rest from time to time during the day. 2 of these machines are old, each having a prob. of $\frac{1}{11}$ of needing rest during the day, and 8 are new, having corresponding prob. of $\frac{1}{21}$.

Assuming that no machine needs rest twice of the same day, Determine the prob. that a particular day

- (i) Just 2 old and no new machine need rest-
- (ii) If just 2 machine need rest, they are of the same type.

Ans:

Let P_1 = prob. that an old machine needs rest = $\frac{1}{11}$

$$\Rightarrow Q_1 = 1 - P_1 = \frac{10}{11}$$

P_2 = prob. that a new machine needs rest = $\frac{1}{21}$

$$Q_2 = \frac{20}{21}$$

then,

$P_1(x=n) = \text{prob. that } n \text{ no. of old machine need rest}$

$$= {}^3C_n \left(\frac{1}{11}\right)^n \cdot \left(\frac{10}{11}\right)^{3-n} \quad \text{--- (v)}$$

and, $P_2(x=n) = \text{prob. that } n \text{ no. of new machine need rest}$

$$= {}^7C_n \left(\frac{1}{21}\right)^n \cdot \left(\frac{20}{21}\right)^{7-n} \quad \text{--- (vi)}$$

(i) the prob. that just 2 machines and no new machine need rest is given by the expression.

$$P_1(x=2) \cap P_2(x=0)$$

$$= P_1(n=2) \cdot P_2(n=0)$$

$$= {}^3C_2 \left(\frac{1}{11}\right)^2 \left(\frac{10}{11}\right)^1 \cdot {}^7C_0 \left(\frac{1}{21}\right)^0 \left(\frac{20}{21}\right)^7$$

from eqn (i) putting $n=2$

from eqn (ii) putting $n=0$

$$= 3 \times \frac{10}{11} \cdot 1 \cdot 1 \cdot \left(\frac{20}{21}\right)^7$$

$$= \frac{30 \times (20)^7}{(11)^3 \times (21)^7} \quad \text{--- (vii)}$$

(ii) Similarly, the prob. that just 2 new machine and no old machine need rest is given by

$$P_1(0) P_2(2) = {}^3C_0 \left(\frac{1}{11}\right)^0 \left(\frac{10}{11}\right)^3 \times {}^7C_2 \left(\frac{1}{21}\right)^2 \left(\frac{20}{21}\right)^5$$

$$= 0.028 \quad \text{--- (viii)}$$

the prob. that if just 2 machines need part, they are of the same type is means that the prob. of either just two old and no new or just 2 new and no old.

Hence the req. prob. is

$$\{P_1(2)P_2(0) \cup P_1(0)P_2(2)\}$$

Since they are disjoint -

$$= P_1(2) \cdot P_2(0) + P_1(0) \cdot P_2(2)$$

$$= {}^7C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^5 + {}^7C_2 \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^5$$

◆ The prob. of a person chitting a target is $\frac{1}{4}$.

(i) If he/she fires 7 times what is the prob. of chitting the target at least twice.

(ii) How many times must he/she fire so that the prob. of hitting the target atleast once is greater than $\frac{2}{3}$.

Ans:-

$$p = \text{prob. of hitting the target} = \frac{1}{4} \quad q = 1 - p = \frac{3}{4}$$

$$q = \frac{3}{4}$$

prob. of

$P(X=n)$ = Prob. of n no. of hits in 7 shots.

$$= {}^7C_n \left(\frac{1}{4}\right)^n \left(\frac{3}{4}\right)^{7-n}$$

(i) Prob. of atleast 2 hits

$$= 1 - \{P(0) + P(1)\}$$

$$= 1 - \left\{ {}^7C_0 \cdot \left(\frac{1}{4}\right)^0 \cdot \left(\frac{3}{4}\right)^7 + {}^7C_1 \cdot \left(\frac{1}{4}\right)^1 \cdot \left(\frac{3}{4}\right)^6 \right\}$$

$$= \frac{4549}{8192}$$

(ii)

prob. of atleast 1 hit with

$$1 - P(0) = 1 - {}^n C_0 \cdot \left(\frac{1}{4}\right)^0 \cdot \left(\frac{3}{4}\right)^n = 1 - \left(\frac{3}{4}\right)^n$$

Q/g

$$1 - \left(\frac{3}{4}\right)^n > \frac{2}{3}$$

$$\Rightarrow -\left(\frac{3}{4}\right)^n > \frac{2}{3} - 1$$

$$\Rightarrow -\left(\frac{3}{4}\right)^n > -\frac{1}{3}$$

$$\Rightarrow \left(\frac{3}{4}\right)^n < \frac{1}{3} \Rightarrow \frac{1}{3} > \left(\frac{3}{4}\right)^n$$

taking log in both side we get,

$$\log\left(\frac{1}{3}\right) > \log\left(\frac{3}{4}\right)^n$$

$$\Rightarrow \log 1 - \log 3 > n \log\left(\frac{3}{4}\right)$$

$$\Rightarrow -\log 3 > n (\log 3 - \log 4)$$

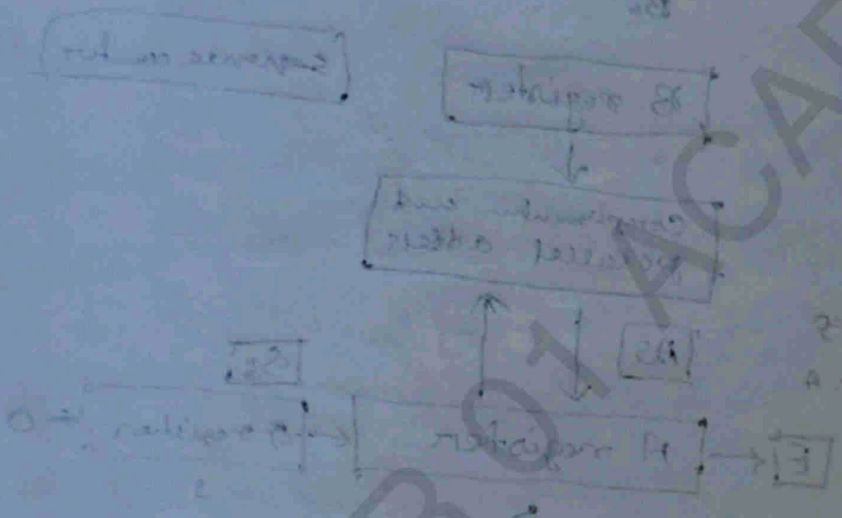
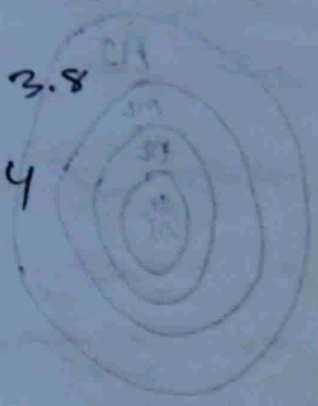
$$\Rightarrow -0.4771 > n (0.4771 - 0.6021)$$

$$\Rightarrow 0.4771 < 0.125n$$

$$\Rightarrow n \leq \frac{6.471}{0.125} = 51.76 \Rightarrow n = 51$$

$$\Rightarrow n \leq 51$$

$$\Rightarrow n = 4$$



ANTOR

16.09.2021

Math

Q In a bombing attack there is a 50% chance that anyone bomb will strike the target. Two direct hits are required to destroy the target completely. How many bombs must be dropped to give a 99% chance or better of completely destroying the target.

Ans:-

Let p = prob. that the bomb will strike the target = $50\% = \frac{1}{2}$

Let n be the number of bombs which should be dropped to ensure 99% chance or better of completely destroying the target. This implies that prob. that out of ' n ' number of bombs, at least two strike the target, is greater than 0.99 .

X be a random number variable representing the no. of bombs striking the target then prob. of

$$P(X=x) = {}^n C_x \cdot p^x \cdot q^{n-x}$$

$$= {}^n C_x \cdot \left(\frac{1}{2}\right)^x \cdot \left(\frac{1}{2}\right)^{n-x} = {}^n C_x \cdot \left(\frac{1}{2}\right)^n \quad \text{--- (1)}$$

~~Assume that~~

$$\text{A/Q } P(X \geq 2) \geq 0.99$$

$$\Rightarrow 1 - P(X \leq 1) \geq 0.99$$

$$\Rightarrow 1 - \{P(X=0) + P(X=1)\} \geq 0.99$$

$$\Rightarrow 1 - \left\{ {}^n C_0 \left(\frac{1}{2}\right)^n + {}^n C_1 \left(\frac{1}{2}\right)^n \right\} \geq 0.99$$

$$\Rightarrow 1 - 0.99 \geq \frac{1}{2^n} + \frac{n}{2^n}$$

$$\Rightarrow 0.01 \geq \frac{1+n}{2^n}$$

$$\Rightarrow 0.01 \cdot 2^n \geq 1+n$$

multiplying b/s by 100

$$2^n \geq 100 + 100n$$

$\Rightarrow$

Destroy the target completely we must drop n no. of bombs such that n will satisfy the (ii).

By trial ~~method~~ ^{method} ~~may not~~ $n = 11$

Hence req. no. of bomb ~~drop~~ will be 11.

§ The mathematical expectation of a ~~rv~~ or expected value of a r.v. Once we had constructed the probability distribution for a random variable, we went to compute the mean or expected value of the r.v. The expected value of a discrete r.v. is a weighted avg. of all possible values of the random variable where the weight are the prob. associated with the corresponding values. The mathematical expression for computing the expected value of a discrete r.v. X with prob. mass funcⁿ is given below.

$$E(X) = \sum_n x f(x) \quad (\text{for discrete r.v.})$$

The mathematical expression for computing the expected value of a continuous random variable X , with prob. density $f(x)$ is given below.

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

(i) Find the mathematical expectation of the numbers on a dice when thrown.

Ans! Let X be the random variable representing the no. on a dice when thrown. Then X can take any one of the values $1, 2, 3, 4, 5, 6$, each with equal probability $\frac{1}{6}$.

$$E(X) = \sum_x x f(x)$$

$$= 1 \cdot P(X=1) + 2 \cdot P(X=2) + \dots + 6 \cdot P(X=6)$$

$$= 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \dots + 6 \cdot \frac{1}{6}$$

$$= \frac{1}{6} (1 + 2 + 3 + 4 + 5 + 6)$$

$$= \frac{7}{2}$$

Hence the req. expectation is $\frac{7}{2}$.

Hence the expectation of a r.v. means, if we repeat for a large no. of times the avg. value of the result is called expectation.

For example in the above problem the exp. is similar to a dice and the result may be $1, 2, 3, \dots, 6$.

In this case the expectation means

with throwing the dice several times and take the avg. of the results. As per the above problem the avg result is $\frac{7}{2}$

Q. Two unbiased dice are thrown. Find the expected values of the sum of numbers of points on them.

Ans:

The prob. function of X is given below:

Value of x	2	3	4	5	6	7	8	9	10	11	12
prob. of $X=x$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Hence the req. expectation will be

$$E(X) = \sum_{n=2}^{12} n \cdot p(n) \\ = 2 \cdot \frac{1}{36} + 3 \cdot \frac{2}{36} + 4 \cdot \frac{3}{36} + \dots + 12 \cdot \frac{1}{36} = 7$$

Q In four process of a coin, let X be the no. of heads. tabulate the 16 possible outcomes with the corresponding values of X . By simple counting, derive the prob. distribution of X , and hence calculate the expected value of X .

Ans:-

let H represent a head, T a tail and X , the random variable denoting the no. of heads.

Sl. No. Outcomes no. of heads

1	HHHH	→ 4
2	HHHT	→ 3
3	HHTH	→ 3
4	HTHH	→ 3
5	THHH	→ 3
6	HHTT	→ 2
7	HTTH	→ 2
8	THTH	→ 2
9	HTHT	→ 2
10	THTH	→ 2
11	THHT	→ 2
12	HTTT	→ 1
13	THTT	→ 1
14	TTHT	→ 1
15	TTTH	→ 1
16	TTTT	→ 0

The random variable X takes the values 0, 1, 2, 3 and 4. Since from the above table we find that the no. of cases favourable to the coming of 0, 1, 2, 3 and 4 heads are 1, 4, 6, 4 and 1 respectively.

Hence,

$$P(X=0) = \frac{1}{16}, P(X=1) = \frac{4}{16}, P(X=2) = \frac{6}{16} = \frac{3}{8}$$

$$P(X=3) = \frac{4}{16} = \frac{1}{4}, P(X=4) = \frac{1}{16}$$

The prob. distribution of X can be summarized as follows.

$X:$	0	1	2	3	4
$P(X):$	$\frac{1}{16}$	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{1}{16}$

Hence,

$$E(X) = \sum_{x=0}^n x P(X=x) = 0 \times \frac{1}{16} + 1 \times \frac{1}{4} + 2 \times \frac{3}{8} + 3 \times \frac{1}{4} + 4 \times \frac{1}{16} = 2$$

Q A coin is tossed until a head appears. what is the expectation of the no. of process required.

Ans: Let X denote the no. of process required to get the 1st head. Then X can happen by the following ways.

Event	x	probability $P(X=u)$
H	1	$\frac{1}{2}$
TH	2	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
TTT	3	$\frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$
TTTH	4	$\frac{1}{8} \times \frac{1}{2} = \frac{1}{16}$
TTTT	5	$\frac{1}{16} \times \frac{1}{2} = \frac{1}{32}$
⋮	⋮	⋮

$$E(x) = \sum_{x=1}^{\infty} x P(X=x)$$

$$= 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + 3 \times \frac{1}{8} + 4 \times \frac{1}{16} + 5 \times \frac{1}{32} + \dots$$

$$\text{Let } S = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{8} + 4 \cdot \frac{1}{16} + 5 \cdot \frac{1}{32} + \dots \quad \text{--- (i)}$$

$$\frac{S}{2} = 0 + \frac{1}{4} + 2 \cdot \frac{1}{8} + 3 \cdot \frac{1}{16} + 4 \cdot \frac{1}{32} + 5 \cdot \frac{1}{64} + \dots \quad \text{--- (ii)}$$

Subtracting (ii) from (i) we obtain

$$S - \frac{S}{2} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \dots$$

$$\Rightarrow \frac{S}{2} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \dots$$

The RHS of the above eqn is a geometric progression series with common ratio $r = \frac{1}{2}$ and

the first term $a = \frac{1}{2}$. Hence $\frac{S}{2} =$

Hence,

$$\frac{s}{2} = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1$$

$$\Rightarrow s = 2$$

Hence the req. mathematical expr

$$E(x) = 2$$

⊠ A box contains 7 white and 3 red balls.

Two balls are drawn together, at random from this box. Compute the prob. that neither of them is white. Find also the prob. of getting one white and one red ball. Hence compute the expected no. of white ball drawn.

Ans:—

Let X denote the no. of white balls drawn.
The prob. distribution of X is given below

$X:$

$$0, 1, 2, \frac{1}{10}, \frac{1}{5}, \frac{1}{10}, \frac{1}{5}, \frac{1}{10}, \frac{1}{5}$$

$$P(X) = \frac{{}^3C_2}{{}^{10}C_2} = \frac{1}{15}, \frac{{}^7C_1 \times {}^3C_1}{{}^{10}C_2} = \frac{7}{15}$$

Since the total no. of balls is 10 so two balls can be drawn in ${}^{10}C_2$ and zero white balls means 2 balls are drawn are red

the prob. that neither of them is white is $\frac{1}{5}$
 the prob. of getting 1 white and 1 red ball is $\frac{9}{15}$

the expected no. of white balls drawn is

$$E(x) = 0 \times \frac{1}{5} + 1 \times \frac{9}{15} + 2 \times \frac{9}{15}$$

$$= \frac{21}{15}$$

Maths

29 Oct 2023

Poisson distribution

Poisson (distribution) was discovered by the French mathematician Denis Poisson. Poisson distribution is a limiting case of the binomial distribution under the following conditions.

(i) n , the no. of trials is very large.

i.e. $n \rightarrow \infty$

(ii) p , the constant probability of success for

each trial is very very small. i.e. $p \rightarrow 0$

(iii) $np = \lambda$ (say) is finite

$$\Rightarrow p = \frac{\lambda}{n}, \quad q = 1 - p = 1 - \frac{\lambda}{n}$$

The probability of 'x' no. of successes in a series of n independent trials is :-

$$b(x, n, p) = \binom{n}{x} p^x q^{n-x}, \quad x=0, 1, 2, \dots, n$$

$$\Rightarrow \lim_{n \rightarrow \infty} b(x, n, p) = \lim_{n \rightarrow \infty} \frac{n!}{(n-x)! x!} \left(\frac{\lambda}{n} \right)^x \left(1 - \frac{\lambda}{n} \right)^{n-x}$$

We know that

$$\lim_{n \rightarrow \infty} n! = \sqrt{2\pi} e^{-n} n^{n+\frac{1}{2}} \quad (\text{Stirling's formula})$$

$$\lim_{n \rightarrow \infty} b(x, n, p) = \lim_{n \rightarrow \infty} \frac{\sqrt{2\pi} e^{-n} n^{n+\frac{1}{2}}}{x! \sqrt{2\pi} e^{-(n-x)} (n-x)^{n-x+\frac{1}{2}}}$$

$$= \frac{\lambda^x}{e^x x!} \lim_{n \rightarrow \infty} \frac{n^{n-x+\frac{1}{2}}}{(n-x)^{n-x+\frac{1}{2}}} \cdot \left(1 - \frac{\lambda}{n} \right)^{n-x}$$

$$= \frac{\lambda^x}{e^x x!} \lim_{n \rightarrow \infty} \frac{\left(1 - \frac{\lambda}{n} \right)^{n-x}}{\left(1 - \frac{\lambda}{n} \right)^{n-x+\frac{1}{2}}}$$

$$= \frac{\lambda^x}{e^x x!} \cdot \frac{\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n} \right)^n \cdot \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n} \right)^{-x}}{\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n} \right)^n \cdot \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n} \right)^{-x+\frac{1}{2}}}$$

We know from limit theory

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n} \right)^n = e^{-\lambda}$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n} \right)^x = 1, \quad \text{if } x \text{ is not a f'n of } n$$

$$\lim_{n \rightarrow \infty} b(n, n, p) = \frac{\lambda^n}{e^\lambda n!} \cdot \frac{e^{-\lambda} \cdot 1}{e^{-n} \cdot 1} = \frac{e^{-\lambda} \cdot \lambda^n}{n!}, \quad n=1, 2, \dots$$

which is the req. poisson distribution formula.

⊗ Show that poisson distribution is the limiting case of binomial distribution.

Ans:—

↑↑

Defⁿ of poisson distribution:—

A random variable X , is said to follow a poisson distribution if it assumes only non-negative values, and its prob. mass function is given by

$$P(X=x) = P(X=x) = \begin{cases} \frac{e^{-\lambda} \cdot \lambda^x}{x!}, & x=0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases} \quad \lambda > 0$$

8) A manufacturer of pins. knows that 5% product are defective, if he sells pins in boxes of hundred and guarantees that not more than 10 pins will be defective, what is the approx prob. that a box will fail to meet the guaranteed quality.

Ans: -

Here $n = 100$

$$p = \text{prob. of defective pin} = 5\% = \frac{5}{100} = 0.05$$

$$\lambda = \text{mean number of defective pins} = np = 100 \times 0.05 = 5$$

Let the random variable X denotes the no. of defective pins in a box of 100. Then by poisson prob. distribution, the prob. of a no. of defective pins in a box is given by:

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

probability that a box will fail to meet the guaranteed quality is

$$P(X > 10) = 1 - P(X \leq 10)$$

$$= 1 - \sum_{x=0}^{10} \frac{e^{-\lambda} \lambda^x}{x!} = 1 - e^{-5} \sum_{x=0}^{10} \frac{5^x}{x!}$$

⑤ A person has two cars which it hires out day by day. The number of demands for a car on each day is distributed as a poisson distribution with (mean) 1.5. Calculate the proportion of days on which (i) neither car is used and (ii) The proportion of days on which some demands is refused.

Ans

Here the random variable X , which denotes the no. of demands for a car on any day follows poisson distribution with mean $\lambda = 1.5$. The proportion of days on which there are X no. of demands for a car is given by

$$P(X=n) = \frac{e^{-\lambda} \cdot \lambda^n}{n!}, \quad n = 0, 1, 2, \dots$$

$$= \frac{e^{-1.5} \cdot (1.5)^n}{n!}, \quad n = 0, 1, 2, \dots$$

(i) Proportion of days on which neither car is used is given by

$$P(X=0) = \frac{e^{-1.5} (1.5)^0}{0!} = e^{-1.5} = 0.2231$$

(ii) Proportion of days on which some demands is refused is :-

$$P(X > 2) = 1 - P(X \leq 2)$$

$$= 1 - \{P(X=0) + P(X=1) + P(X=2)\}$$

$$= 1 - \left\{ \frac{e^{-1.5} \cdot (1.5)^0}{0!} + \frac{e^{-1.5} \cdot (1.5)^1}{1!} + \frac{e^{-1.5} \cdot (1.5)^2}{2!} \right\}$$

=

Q A manufacturer, who produces medicine bottles finds that 0.1% of the bottles are defective. The bottles are packed in boxes containing five hundred bottles. A drug manufacturer buys 100 box from the producer. Using poisson distribution find how many boxes are.

(i) No defective

(ii) at least two defective.

Ans:

$$\text{Here } N = 100, n = 500$$

$$P = \text{prob. of a defective bottle} = \frac{0.1}{100} = 0.001$$

$$\lambda = np = 500 \times 0.001 = 0.5$$

Let the random variable, X , denote the no. of defective bottles in a box of 500. Then by poisson distribution, the probability of x no. of defective bottles in a box is given by

$$P(X=x) = \frac{e^{-0.5} (0.5)^x}{x!}$$

$$= \frac{0.6065 \times (0.5)^x}{x!} \quad ; x=0, 1, 2, \dots$$

Hence the consignment of 100 boxes the no. of containing x no. of defective bottles is

$$f(x) = NP(X=x) = 100 \times \frac{0.6065 \times (0.5)^x}{x!}$$

(i) No. of boxes containing no defective bottle is:-

$$f(0) = 100 \times 0.6065$$

$$f(0) = 100 \times \frac{0.6065 \times (0.5)^0}{0!} = 0.6065 \times 100 = 60.65 \approx 61 \text{ boxes.}$$

(ii) No. of boxes containing atleast two defective bottles:-

$$= 100 \times P(X \geq 2) = 100 \times \{1 - P(X \leq 1)\}$$

$$= 100 \times \{1 - \{P(X=0) + P(X=1)\}\}$$

$$= 100 \times \left\{ 1 - \left[\frac{100 \times 0.6065 \times (0.5)^0}{0!} + \frac{0.6065 \times 0.5}{1!} \right] \right\}$$

$$\approx 9$$

⑧ 6 ~~no.~~ no. of heads

⑨ Six coin of tossed 6400 times. using the poisson distribution, find the approximate probability of getting 6 heads 20 times.

Ans:

The prob. of obtaining 6 no. of heads in 1 throw of six coin is $p = \frac{1}{2^6} = \frac{1}{64}$

$$\therefore \text{Here, } \lambda = np = 6400 \times \frac{1}{64} = 100$$

Hence by using poisson distribution, the req. prob. of getting 6 no. of heads 20 times are given by,

$$P(X=20) = \frac{e^{-\lambda} \lambda^{20}}{20!} = \frac{e^{-100} (100)^{20}}{20!}$$

Q In a book of 520 pages, 390 typing errors occur. Assuming poisson distribution for the no. of error per page, find the prob that a random sample of 5 pages will contain no error.

Ans

The avg. no of typing mistake per page in book is given by $\lambda = \frac{390}{520} = 0.75$

Hence by using poisson distribution, the prob. of x no. of error per page is given by

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \quad ; \quad x=0,1,2,\dots$$

The req. prob. that a random sample of 5 pages will contain no. error is given by :-

$$\begin{aligned} P(X=0) &= \left\{ \frac{e^{-0.75} \cdot (0.75)^0}{0!} \right\}^5 \\ &= \left\{ e^{-0.75} \right\}^5 \\ &= e^{-3.75} \end{aligned}$$

Two dimensional random variable.

Let x and y be two random variables defined in the same sample space S , then the function (x, y) that assigns a point in $\mathbb{R}^2$ is called a two dimensional random variable.

Two dimensional or Joint probability mass function

Let x and y be random variables on a sample space ' S ' with respective image sets

$$x(S) = \{x_1, x_2, x_3, \dots, x_n\} \text{ and } y(S) = \{y_1, y_2, y_3, \dots, y_m\}$$

We make the product sets

$$X(S) \times Y(S) = \{x_1, x_2, x_3, \dots, x_n\} \times \{y_1, y_2, y_3, \dots, y_m\}$$

into a probability of the space by defining the probability of the ordered pair (x_i, y_j) to be

$$P(X = x_i, Y = y_j)$$

the function P on $X(S) \times Y(S)$ defined by

$P_{ij} = P(X = x_i, Y = y_j)$ is called the joint probability mass function of x and y .

the joint prob. ~~mass~~ function is usually represented by tabular form as given below:-

$$p_{ij} = P(X=x_i, Y=y_j)$$

$Y \backslash X$	y_1	y_2	y_3	$\dots$	y_n	Total
x_1	p_{11}	p_{12}	p_{13}	$\dots$	p_{1n}	$p_{1\cdot}$
x_2	p_{21}	p_{22}	p_{23}	$\dots$	p_{2n}	$p_{2\cdot}$
x_3	p_{31}	p_{32}	p_{33}	$\dots$	p_{3n}	$p_{3\cdot}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_n	p_{n1}	p_{n2}	p_{n3}	$\dots$	p_{nn}	$p_{n\cdot}$
Total	$p_{\cdot 1}$	$p_{\cdot 2}$	$p_{\cdot 3}$	$\dots$	$p_{\cdot n}$	1

Defⁿ:- If (X, Y) is a two dimensional random variable then the joint density f^n of X, Y also known as the joint prob. mass function X, Y denoted by $p_{X,Y}$ is defined as

$$p_{X,Y}(x_i, y_j) = \begin{cases} P(X=x_i, Y=y_j) & \text{for } (x_i, y_j) \in X(s) \times Y(s) \\ 0, & \text{otherwise} \end{cases}$$

Marginal probability function:-

Let (X, Y) be a discrete two-dimensional r.v. which takes up countable no. of values (x_i, y_j) .

Then the prob. distribution of X is determined as follows:-

$$P_x(x_i) = P(X=x_i)$$

$$= P(X=x_i \cap Y=y_1) + P(X=x_i \cap Y=y_2) + \dots + P(X=x_i \cap Y=y_m)$$

$$= P_{i1} + P_{i2} + P_{i3} + \dots + P_{im}$$

$$= \sum_{j=1}^m P_{ij} = P_{i\cdot}$$

and is known as marginal probability mass function or marginal density function of X .

~~Also~~ $\sum$

$$\text{Also } \sum_{i=1}^n P_{i\cdot} = 1$$

Similarly we can define the marginal p.m.f of Y also.

Defⁿ:- Let (X, Y) be a discrete two dimensional r.v. then the conditional density function or the conditional probability mass fⁿ of X ; given $Y=y$ denoted by

$f_{X|Y}(x/y)$ is defined as,

$$P_{X|Y}(x/y) = \frac{P(X=x, Y=y)}{P(Y=y)}, \quad \text{provided } P(Y=y) \neq 0.$$

Two dimensional distribution function:-

Defⁿ:- the distribution function of the two dimensional random variable (X, Y) is a real valued function F defined for all $x \in \mathbb{R}, y \in \mathbb{R}$, by the relation

$$F_{xy}(x, y) = P(X \leq x, Y \leq y)$$

④ The joint probability distribution of two random variables X and Y :

$$P(X=0, Y=1) = \frac{1}{3},$$

$$P(X=1, Y=-1) = \frac{1}{3} \text{ and } P(X=1, Y=1) = \frac{1}{3}$$

find (i) marginal distribution of X and Y . (ii) the conditional probability distribution of X given $Y=1$.

Ans From the given data the joint prob. distribution of X and Y is given below.

$X \backslash Y$	0	1	-1	marginal dist ⁿ of Y
0	0	0	0	0
1	$\frac{1}{3}$	0	$\frac{1}{3}$	$\frac{2}{3}$
-1	0	$\frac{1}{3}$	0	$\frac{1}{3}$
marginal dist ⁿ of X	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	1

values of x :-

$p(x=x)$:- $\frac{1}{3}$ $\frac{1}{3}$ $\frac{1}{3}$

value of y :-

$p(y=y)$:- 0 $\frac{2}{3}$ $\frac{1}{3}$

(ii) the conditional p.d. of x given $y=1$ is

$$p(x=x | y=1) = \frac{p(x=x, y=1)}{p(y=1)}$$

$$p(x=-1 | y=1) = \frac{p(x=-1, y=1)}{p(y=1)} = \frac{0}{\frac{2}{3}} = 0$$

$$p(x=0 | y=1) =$$

$$p(x=1 | y=1) =$$

⊠ for the following bivariate probability distribution of x and y find.

$y \backslash x$	1	-2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

(i) $P(X \leq 1; Y=2)$ (ii) $P(X \leq 1)$ (iii) $P(Y \leq 3)$ and

(iv) $P(X \leq 3, Y \leq 4)$

The marginal distributions are given below -

$X \backslash Y$	1	2	3	4	5	6	$P_X(m)$
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	$\frac{8}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{10}{16}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$	$\frac{8}{64}$
$P_Y(y)$	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{6}{32}$	$\frac{16}{64}$	1

$$\begin{aligned} (i) \quad P(X \leq 1, Y=2) \\ = P(X=0, Y=2) + P(X=1, Y=2) \\ = 0 + \frac{1}{16} = \frac{1}{16} \end{aligned}$$

$$\begin{aligned} (ii) \quad P(X \leq 1) &= P(X=0) + P(X=1) \\ &= \frac{8}{32} + \frac{16}{32} = \frac{24}{32} \end{aligned}$$

$$\begin{aligned} (iii) \quad P(Y \leq 3) &= P(Y=1) + P(Y=2) + P(Y=3) \\ &= \frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{23}{64} \end{aligned}$$

~~(iv) $P(X \leq 3, Y \leq 4)$~~

$$(iv) P(X \leq 3, Y \leq 4)$$

$$= P(X=0, Y \leq 4) + P(X=1, Y \leq 4)$$

$$= \frac{3}{32} + \frac{6}{16} + \frac{6}{64} = \frac{6+24+6}{64}$$

$$= \frac{36}{64}$$

maths

18 Nov 2022

Q The joint prob. distribution of two random variable X and Y given below :-

$Y \backslash X$	1	2	3	4
1	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$
2	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{2}{36}$
3	$\frac{5}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
4	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{5}{36}$

Find (i) The marginal distributions of X and Y
 (ii) Conditional distribution of X , given the value of $Y=1$ and that of Y given the value of $X=2$.

Ans:- The marginal distribution of x is defined as

$$P(X=x) = \sum_y P(X=x, Y=y)$$

$$\begin{aligned}\text{Hence, } P(X=1) &= \sum_y P(X=1, Y=y) \\ &= P(X=1, Y=1) + P(X=1, Y=2) + P(X=1, Y=3) + P(X=1, Y=4) \\ &= \frac{10}{36}\end{aligned}$$

$$\begin{aligned}P(X=2) &= P(X=2, Y=1) + P(X=2, Y=2) + P(X=2, Y=3) + P(X=2, Y=4) \\ &= \frac{9}{36}\end{aligned}$$

$$P(X=3) = \frac{8}{36}$$

$$P(X=4) = \frac{9}{36}$$

the marginal distribution of y is defined as:

$$P(Y=y) = \sum_x P(X=x, Y=y)$$

$$\begin{aligned}P(Y=1) &= \sum_x P(X=x, Y=1) \\ &= P(X=1, Y=1) + P(X=2, Y=1) + P(X=3, Y=1) + P(X=4, Y=1) \\ &= \frac{11}{36}\end{aligned}$$

$$\begin{aligned}P(Y=2) &= P(X=1, Y=2) + P(X=2, Y=2) + P(X=3, Y=2) + P(X=4, Y=2) \\ &= \frac{9}{36}\end{aligned}$$

$$P(Y=3) = \frac{7}{36}$$

$$P(Y=4) = \frac{9}{36}$$

(ii) The conditional prob. f^x of X given Y is defined as follows :-

$$P(X=x | Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)}$$

$$\therefore P(X=1 | Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{4/36}{11/36} = \frac{4}{11}$$

$$P(X=2 | Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)} = \frac{1/36}{11/36} = \frac{1}{11}$$

$$P(X=3 | Y=1) = \frac{5}{11} \quad \text{and} \quad P(X=4 | Y=1) = \frac{1}{11}$$

Hence the conditional distribution of X given $Y=1$ is :-

X	1	2	3	4
$P(X=x Y=1)$	$\frac{4}{11}$	$\frac{1}{11}$	$\frac{5}{11}$	$\frac{1}{11}$

Similarly, we can obtain the conditional distribution of Y for $X=2$ as given below :-

Y	1	2	3	4
$P(Y=y X=2)$	$\frac{1}{9}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{9}$

Q A two dimensional r.v X, Y have the following joint distribution:

$$P(X=x, Y=y) = \frac{x^2+y}{32}, \text{ for } x=0,1,2,3 \text{ and } y=0,1$$

Find the marginal distribution of X and Y .

Ans:-

The joint distribution table is given below.

$x \backslash y$	0	1	2	3
0	0	$\frac{1}{32}$	$\frac{4}{32}$	$\frac{9}{32}$
1	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{5}{32}$	$\frac{10}{32}$

Q Two dimensional r.v X, Y have joint prob. mass function $P(x, y) = \frac{1}{27} (2x+y)$, where x and y can assume only the integer value 0, 1 and 2.

(i) Find the marginal distribution of X and Y .

(ii) Find the conditional dist of Y for $X=x$.

Ans

Karl Pearson coefficient of correlation:-

As a measure of intensity or degree of linear relationship b/w two variables Karl Pearson, a British mathematician develop a formula called correlation coefficient b/w two random variable x and y usually denoted by $r(x, y)$ or simply r_{xy} , ~~is~~ is a numerical measure of linear relationship b/w them and defined as

$$r(x, y) = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

where,

$$\text{Cov}(x, y) = E\{(x - E(x))(y - E(y))\} = \frac{1}{n} \sum x_i y_i - \bar{x} \bar{y}$$

$$\sigma_x^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2$$

$$\sigma_y^2 = \frac{1}{n} \sum y_i^2 - \bar{y}^2$$

Thm:- Correlation coefficient is independent of change of origin and scale.

$$\text{let } U = \frac{x-a}{h}, \quad V = \frac{y-b}{k}$$

$$\Rightarrow x = a + hU \quad y = b + kV$$

where a, b, h, k are constant and $h > 0, k > 0$.

We need to prove that $r(x, y) = r(u, v)$

Since $x = a + hU$ and $y = b + kV$, on taking expectation, we get,

$$E(x) = a + h E(u) \quad \text{and}$$

$$E(y) = b + k E(v)$$

$$\text{Now, } x - E(x) = a + h v - a - h E(u) = h(v - E(u))$$

$$y - E(y) = b + k v - b - k E(v) = k(v - E(v))$$

$$\text{Cov}(x, y) = E\{(x - E(x))(y - E(y))\}$$

$$= E\{h(v - E(u))k(v - E(v))\}$$

$$= hk E\{(v - E(u))(v - E(v))\}$$

$$= hk \text{Cov}(u, v)$$

$$\text{Now, } \sigma_x^2 = E\{(x - E(x))^2\} = E\{h^2(v - E(u))^2\}$$

$$= h^2 E\{(v - E(u))^2\}$$

$$= h^2 \sigma_u^2$$

$$\Rightarrow \sigma_x = h \sigma_u$$

$$\Rightarrow \sigma_x = h \sigma_u$$

Similarly,

$$\sigma_y = k \sigma_v$$

$$\therefore r(x, y) = \frac{\text{Cov}(x, y)}{\sigma_x \cdot \sigma_y}$$

$$= \frac{hk \text{Cov}(u, v)}{h \sigma_u \cdot k \sigma_v}$$

$$= \frac{\text{Cov}(u, v)}{\sigma_u \cdot \sigma_v}$$

$$= r(u, v)$$

Hence the correlation coefficient is independent of change of origin and scale.

⑧ A computer while computing calculating correlation coefficient b/w two variables X and Y from 25 pairs of observations obtained the following results :-

$n = 25$, $\Sigma x = 125$, $\Sigma x^2 = 650$, $\Sigma y = 100$, $\Sigma y^2 = 460$, $\Sigma xy = 508$,
It was however later discovered at the time of checking that he copied down two pairs as

X	Y
6	14
8	6

, while the correct values were

X	Y
8	12
6	8

Obtain the correct value of correlation coefficient.

Ans:-

$$\text{Corrected } \Sigma x = 125 - 6 - 8 + 8 + 6 = 125$$

$$\text{Corrected } \Sigma y = 100 - 14 - 6 + 12 + 8 = 100$$

$$\text{Corrected } \Sigma x^2 = 650 - 6^2 - 8^2 + 8^2 + 6^2 = 650$$

$$\text{Corrected } \Sigma xy = 508 - 6 \times 14 - 8 \times 6 + 8 \times 12 + 6 \times 8 = 520$$

$$\text{Corrected } \Sigma y^2 = 460 - 14^2 - 6^2 + 12^2 + 8^2 = 436$$

$$\bar{X} = \frac{1}{n} \Sigma x = \frac{1}{25} \times 125 = \frac{1}{25} \times 125 = 5$$

$$\bar{Y} = \frac{1}{n} \Sigma y = \frac{1}{25} \times 100 = 4$$

$$\begin{aligned} \text{cov}(x, y) &= \frac{1}{n} \Sigma xy - \bar{X} \bar{Y} \\ &= \frac{1}{25} \times 520 - 5 \times 4 \\ &= \frac{4}{5} \end{aligned}$$

$$\sigma_x^2 = \frac{1}{n} \sum x^2 - \bar{x}^2$$

$$= \frac{1}{25} \times 650 - 5^2$$

$$= 1$$

$$\sigma_y^2 = \frac{1}{n} \sum y^2 - \bar{y}^2 = \frac{1}{25} \times 436 - 16 = \frac{36}{25}$$

corrected $r(x, y) = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$

$$= \frac{45}{1 \times 5} = 9$$

$$= \frac{45}{1 \times \frac{6}{5}} = \frac{12}{3}$$

x	y
1	5
2	3